



UNIVERSITY of the
WESTERN CAPE



Emerging Trends in Information Systems

IFS315/IFS354: Exercise 3

2025

Marks	50
Due date	17 th May 2025 – 19:00
Submission	Via iKamva and Google Drive
Lecturer	Ruchen Wyngaard
Department	Information Systems

OVERVIEW

For this exercise, you will develop a machine learning-based vision system to solve one of two real-world classification problems: Smart Waste Sorting or Crop Identification. This project will introduce you to applying computer vision and machine learning techniques to practical classification challenges.

Vision systems are a critical component of many modern technologies, from autonomous vehicles to quality control in manufacturing. By completing this exercise, you'll develop foundational skills in creating a machine learning based vision system and its potential application to address real-world problems. You'll develop your own machine learning model to test and refine your system, experiencing firsthand the practical considerations involved in implementing computer vision for waste management or agricultural applications.

OBJECTIVES

You are required to develop a machine learning model suitable for a vision system, this includes:

- Collect and pre-process image data for machine learning
- Develop and train a machine learning model for image classification
- Evaluate the performance of your vision system
- Discuss the real-world applications of your vision system
- Explore the ethical considerations of discussed real-world applications

PROJECT OPTIONS

Choose one of the following vision system projects to implement:

Topic	Description
Smart Waste Sorting System	Develop a vision system that classifies waste items into categories such as plastic, paper, metal, glass, and organic material. Your model must recognize objects from various angles while addressing challenges like partially obscured items and similar-looking materials with different disposal requirements.
Crop Identification System	Create a vision-based solution that identifies and classifies different crop types and varieties from images. Your model should distinguish between crop species despite variations in growth stages, health conditions, and environmental factors such as soil backgrounds and partial visibility.

REQUIREMENTS

Data Collection and Pre-processing

- Collect a dataset of images for each class/object in your chosen project
- Pre-process the images (e.g., resizing, normalization)
- Split the data into training, validation, and test sets

Model Development

- Choose an appropriate machine learning framework and neural network architecture
- Using Python, implement and train your model using the framework of your choice (e.g., TensorFlow, PyTorch)
- Apply relevant techniques to improve model performance (e.g., data augmentation, transfer learning)

Evaluation

- Evaluate your model using appropriate metrics (e.g. accuracy, precision, recall)
- Analyze the model's performance, including its strengths and limitations
- Test the model on new, unseen data to assess its generalization

Documentation and Reporting

- Provide a clear explanation of your development methodology
- Discuss the potential real-world applications, limitations and ethics of your system
- Propose methods to ensure responsible use of your vision system

DELIVERABLES

- Source code (well-commented and organized)
- Final trained model
- Project report (PDF, 3-7 pages excluding references) covering all aspects of your project
- 5 to 10-minute video presentation demonstrating your vision system in action **(It is recommended that you make use of your web cam for the demonstration)**

REPORT FORMAT

- Follow formal report structure.
- Font Type: Arial Font Size: 11; 1.15 Line Spacing with margins of 2.54 CMs on all sides of pages.
- Appendices are placed at the back of the report and each appendix should start on a new page and have a letter assigned to it, making it easy to refer to in your report e.g. Appendix A, Appendix B, etc.

- Also attach an Appendix with all your references (Harvard format).

POINTS TO NOTE

- **This is an individual exercise**
- All completed exercises should be submitted in electronic format via iKamva
- Your video presentation must be submitted via Google Drive
- Hand-written work will not be marked
- No late submissions will be accepted